

Experiment 12: Port Scanning Tools in Kali Linux

Aim

To execute Nmap port scanning commands in Kali Linux to identify open ports, scan a range of ports, and perform TCP SYN, TCP connect, and UDP scans.

Procedure

1. Start the Kali Linux operating system.
2. Open the Terminal application.
3. Check whether Nmap is installed by entering `nmap --version`.
4. Use `127.0.0.1` as the target IP address, which refers to the local machine.
5. Execute each Nmap command separately.
6. Observe the output for each command.
7. Record the port numbers, port states, and associated services.

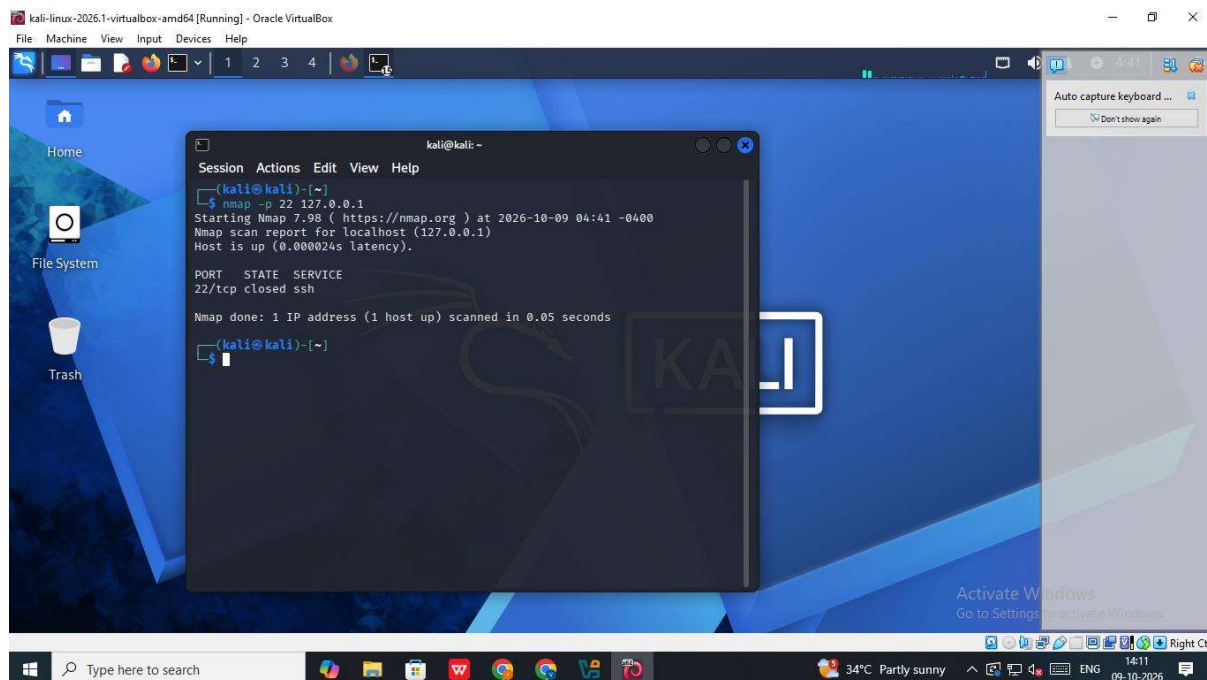
Input and Output

Input 1: Scan a specific port

Command:

```
nmap -p 22 127.0.0.1
```

Output 1:

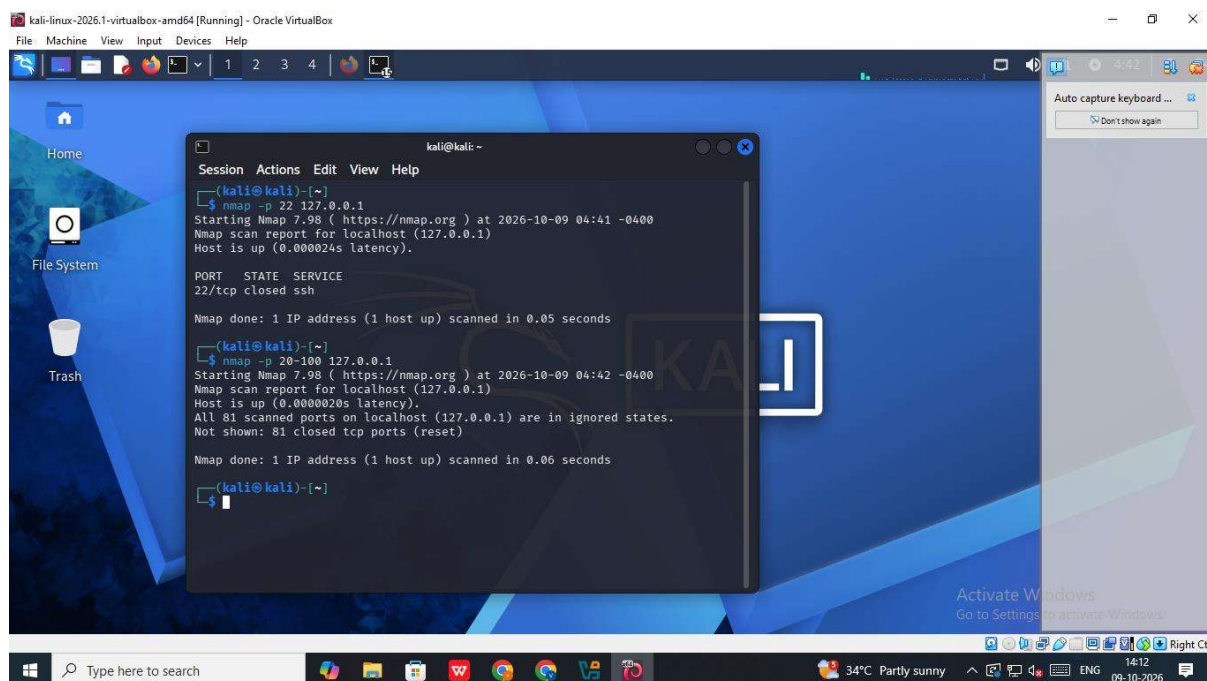


Input 2: Scan a range of ports

Command:

```
nmap -p 20-100 127.0.0.1
```

Output 2:

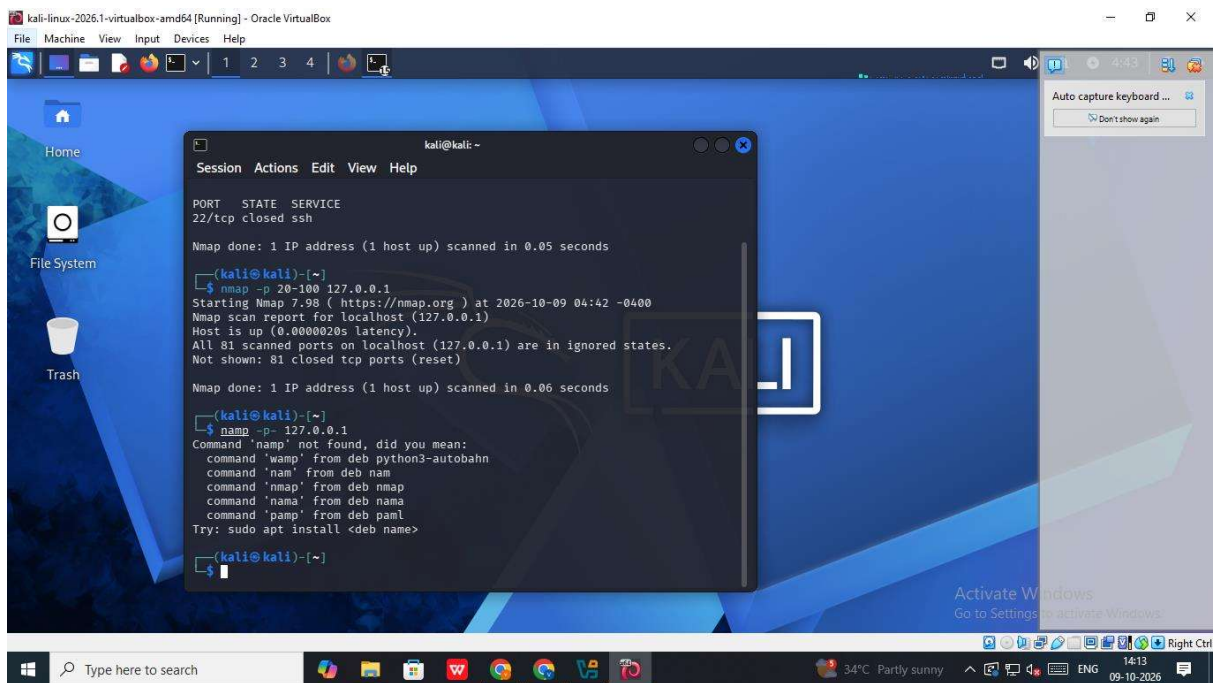


Input 3: Scan all TCP ports

Command:

```
nmap -p- 127.0.0.1
```

Output 3:



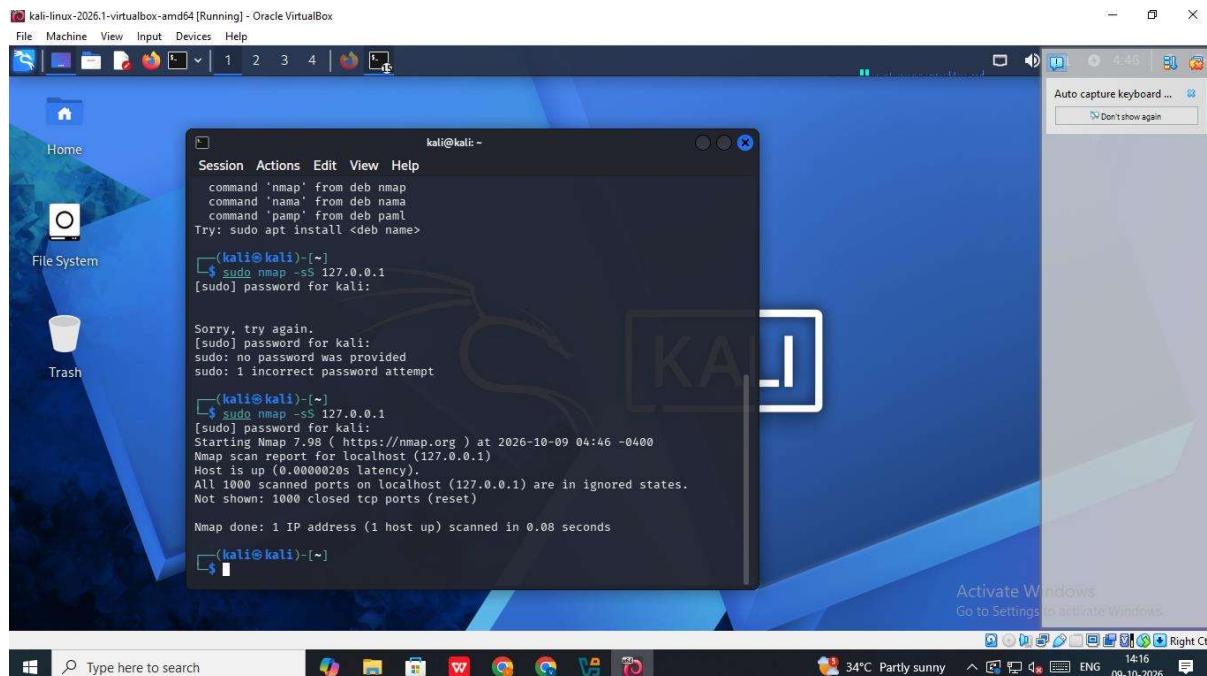
```
kali@kali: ~  
Session Actions Edit View Help  
PORT STATE SERVICE  
22/tcp closed ssh  
Nmap done: 1 IP address (1 host up) scanned in 0.05 seconds  
kali@kali: ~  
$ nmap -p 20-100 127.0.0.1  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-10-09 04:42 -0400  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.0000020s latency).  
All 81 scanned ports on localhost (127.0.0.1) are in ignored states.  
Not shown: 81 closed tcp ports (reset)  
Nmap done: 1 IP address (1 host up) scanned in 0.06 seconds  
kali@kali: ~  
$ nmap -p- 127.0.0.1  
Command 'nmap' not found, did you mean:  
command 'wamp' from deb python3-autobahn  
command 'nam' from deb nam  
command 'nmap' from deb nmap  
command 'nama' from deb nama  
command 'pamp' from deb paml  
Try: sudo apt install <deb name>  
kali@kali: ~  
$
```

Input 4: TCP SYN scan

Command:

```
sudo nmap -sS 127.0.0.1
```

Output 4:

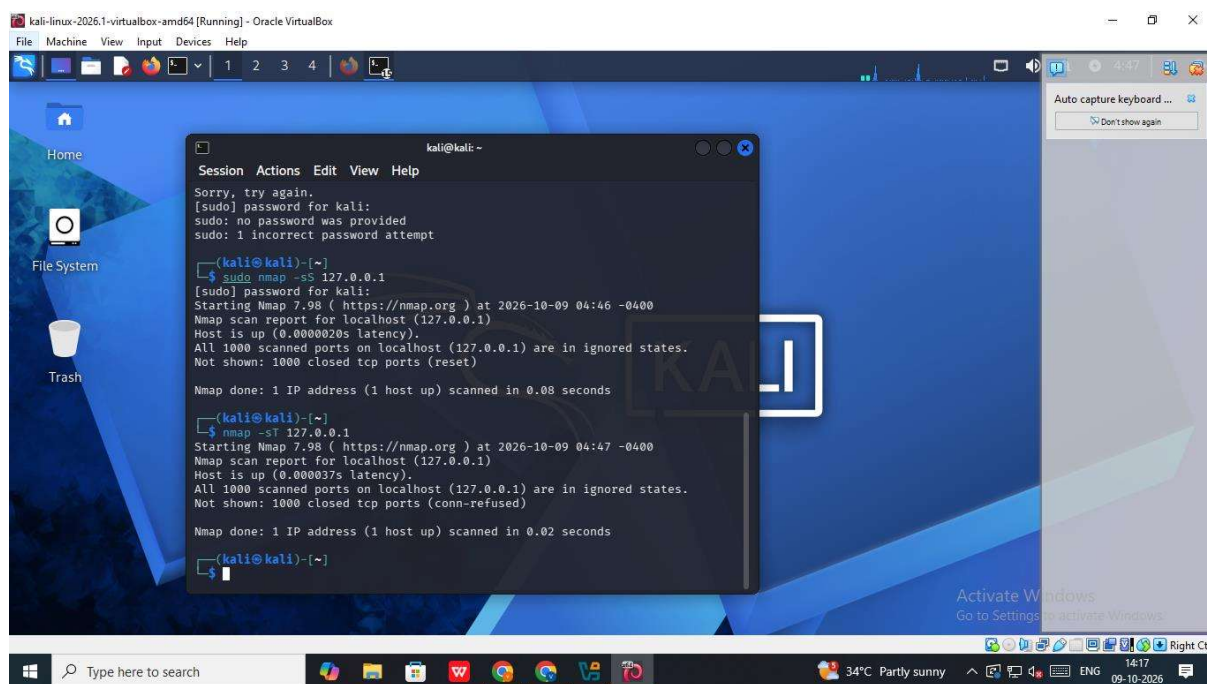


Input 5: TCP connect scan

Command:

```
nmap -sT 127.0.0.1
```

Output 5:

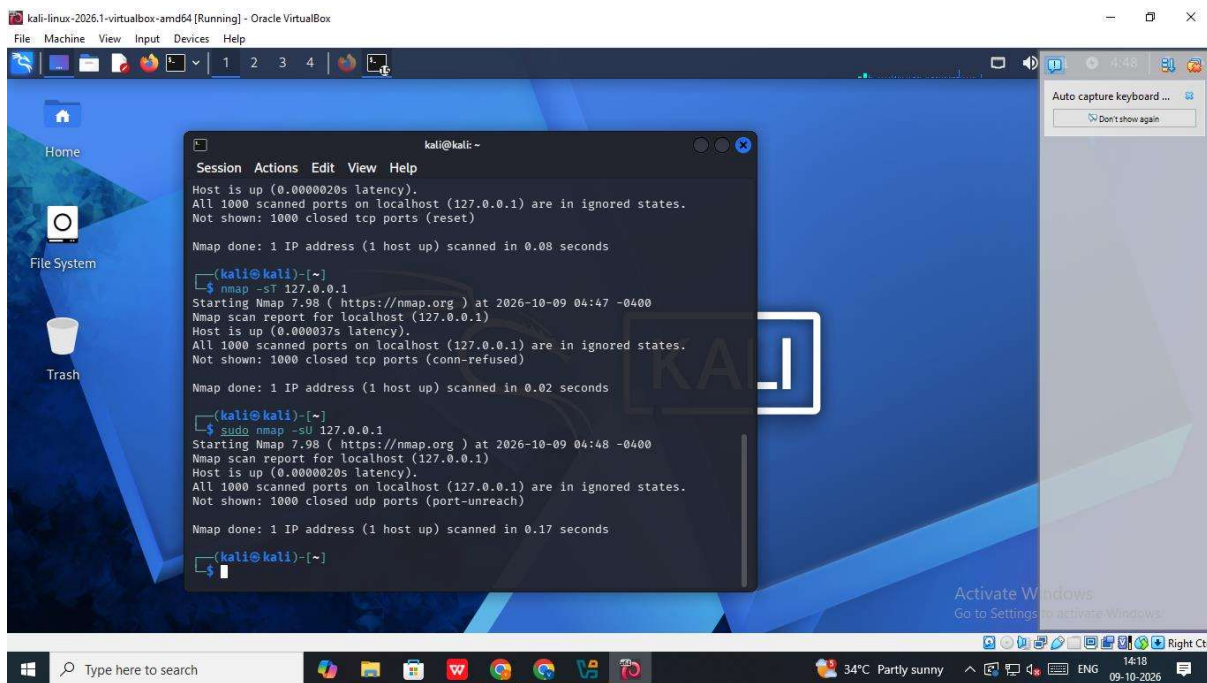


Input 6: UDP scan

Command:

```
sudo nmap -sU 127.0.0.1
```

Output 6:



```
kali@kali: ~  
Session Actions Edit View Help  
Host is up (0.000020s latency).  
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.08 seconds  
  
kali@kali:~$ nmap -sT 127.0.0.1  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-10-09 04:47 -0400  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.000037s latency).  
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.  
Not shown: 1000 closed tcp ports (conn-refused)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.02 seconds  
  
kali@kali:~$ sudo nmap -sU 127.0.0.1  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-10-09 04:48 -0400  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.000020s latency).  
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.  
Not shown: 1000 closed udp ports (port-unreach)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.17 seconds  
  
kali@kali:~$
```

Result

The Nmap port scanning commands were studied and demonstrated in Kali Linux. The experiment covered specific-port scanning, port-range scanning, all-TCP-port scanning, TCP SYN scanning, TCP connect scanning, and UDP scanning. These commands help identify port states and investigate network services on authorized systems.